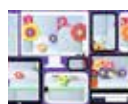




Contactless neighbours

Nextdoor, an app to connect neighbours is seeing a spike in users while also ensuring social distancing. <https://digit.in/apr20-31>



Tech for the dead

The death industry has been forced to adapt to lockdowns with livestreamed funerals. <https://digit.in/apr20-32>

complications. The ones infected can unknowingly spread the virus to others while their liver is gradually reaching the stage of cirrhosis or liver cancer, in many cases.

The CDC has stated that Hepatitis C is a leading cause of liver disease, liver cancer, and liver transplants and it is 10 times more infectious than HIV. It is also no surprise that the relatively unnoticeable nature of hepatitis viruses have caused them to become the 7th leading cause of death in the world. In 2016, the number of deaths worldwide from viral hepatitis touched 1.4 million, beating the death count of tuberculosis, malaria, and HIV individually. Nevertheless, the silent nature of the viruses have understated their impact and therefore, hepatitis has never seen as much public policy and intervention around it, as compared to say, HIV. Although Hepatitis B has had a vaccine for many years, researchers haven't yet developed one for Hepatitis C. Twenty potential vaccines have been studied in animals, however, a few of these have actually undergone testing in humans. Nevertheless, as per the CDC, new treatments have been developed and successfully used to cure the disease, and these treatments are also cost-effective.

FUNGAL INFECTIONS

While most people think of fungal infections as relatively non-threatening infections, there are some very serious and often life-threatening forms of fungal diseases. Many of these are relatively unknown to the common man. GAFFI (Global Action Fund for Fungal Infections) estimated that these fungal diseases claim about 1.5 to 2 million lives every year on a global scale. Moreover, we're seeing new drug-resistant fungi such as the *Candida auris* fungus. This novel fungus spread rapidly and caused massive outbreaks in hospitals across the globe. The fungus targets people with compromised immune systems such as people living with HIV/AIDs, asthma, cancer, critically-ill people, premature babies, and others. The *C. auris* is extremely tenacious and is immune to major antifungal medications which

makes it one of the most contractible health threats.

This new fungal infection is one of the deadliest silent killers today since fungal diseases are firstly, extremely hard to diagnose given the unavailability of laboratory tests and because they are commonly mistaken for Tuberculosis. Additionally, such hospital-acquired diseases are difficult to diagnose yet again since the patient is already sick. The underreporting could also be due to fungal infections not being reportable by law. The fungus was reported for the first time in the US in 2013 and a total of 998 cases have been reported since. However, these are only the reported numbers. Estimates say that a major number of deaths may have gone undetected altogether.



A strain of *Candida auris* fungus cultured in a petri dish at a CDC laboratory

MEASLES

Measles outbreaks saw a massive resurgence in the United States in 2019. Other countries such as Ukraine, Philippines, Venezuela, Brazil, Italy, France and Japan also felt the brunt of this returning disease. Measles is an extremely contagious respiratory disease which is caused by a viral infection from the paramyxovirus family. While this dis-

ease was extremely common and deadly at one time, measles can almost always be prevented by a simple vaccine. Public health professionals had even declared measles as being controlled over two decades ago, however, the cases started escalating again in 2019.

The first six months of 2019 produced more cases of measles than any year since 2006, as per WHO. A whopping 364,808 measles cases were reported by about 182 countries. The cases increased 10-fold in Africa, twofold in European regions and threefold in the Western Pacific region. Why this sudden spike? It has been noticed that most of the countries which were reporting the outbreak have very low vaccination rates. The anti-vaxxers' community possibly refused to get their measles shots which caused the alarming increase in measles cases. However, WHO noticed that even countries with high vaccine coverage were seeing increased measles activity. This is most likely due to the drop in herd immunity. When a large percentage of the population is vaccinated against a certain disease, it is significantly harder for infectious diseases to proliferate. However, since measles is an extremely contagious disease, herd immunity can only be achieved when a high level of the population is immune to the virus. According to a study by The University of Minnesota, at least a 95 percent community vaccination rate is required to prevent sustained transmission. Sadly, as per WHO, only 86 percent of children received their first dose of measles vaccine globally while merely 69 percent received the second. Ultimately, it comes to mitigation of misinformation, educating those behind the anti-vaxx movements and implementing effective public health policies. **4**

THE NOT-SO-SILENT KILLER

What makes COVID-19 more threatening than SARS and MERS?

The SARS outbreak in 2003 killed over 858 people and showed a mortality rate of 9.63% while MERS (Middle East respiratory syndrome) showed a mortality rate of a shocking 34.45%. COVID-19's mortality rate is estimated to be much lower (as of now), however, it is exponentially more infectious than SARS and MERS, which increases its overall lethality count by a large margin.